

Avatar Schema Documentation

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Overview

The Avatar Schema in Raritone is designed to store AI-generated avatar information for virtual try-on experiences.

The avatar system helps create personalized virtual representations of users based on:

- Body measurements
- Body type
- Skin tone
- Gender
- Uploaded images

This schema prepares the backend for future AI-powered avatar generation and virtual fitting workflows.

Purpose of Avatar Schema

The Avatar Schema is used for:

- AI virtual try-on systems
- Personalized clothing visualization
- Recommendation engine support
- Body compatibility analysis
- Future AR/VR integration

Avatar Workflow

User Uploads Image



Backend Receives User Data



AI Avatar Processing Starts



Avatar Generated



Avatar Stored In Database



Used For Virtual Try-On

Schema Fields

Example Structure

```
{  
  userId,  
  avatarImage,  
  bodyType,  
  skinTone,  
  gender,  
  avatarModel,  
  createdAt  
}
```

Field Explanations

Field	Purpose
userId	Links avatar to user
avatarImage	Stores avatar image URL
bodyType	Stores body structure type
skinTone	Stores user skin tone
gender	Stores gender information
avatarModel	Stores AI avatar model reference
createdAt	Stores creation timestamp

Relationships

The Avatar Schema is connected with:

- User schema
- TryOn schema

- ProductMapping schema

This enables:

- Personalized recommendations
- Virtual fitting
- AI compatibility matching

AI Usage

The avatar system supports future AI features such as:

- AI avatar generation
- Virtual clothing fitting
- Body compatibility analysis
- Personalized styling recommendations

Backend Integration

The Avatar Schema can be integrated with:

- Try-on APIs
- AI processing services
- Realtime Socket.IO updates
- Recommendation engines

Possible APIs

POST /api/avatar/generate

GET /api/avatar/:id

DELETE /api/avatar/:id

Realtime Workflow Support

Using Socket.IO, users can receive:

- Realtime avatar generation updates
- Processing status notifications
- Instant avatar delivery

Security Considerations

The avatar workflow includes:

- JWT protection
- Joi validation
- Secure upload handling
- Protected APIs

This ensures:

- Safe user data handling
- Secure image processing
- Controlled access

Scalability Considerations

For large-scale systems:

- Avatars can be stored using CDN services
- Image compression can reduce bandwidth
- Redis caching may improve retrieval speed
- Queue systems can support AI processing

Future Improvements

Possible future enhancements:

- 3D avatar generation
- AR/VR avatar integration
- Realtime body scanning
- AI body analysis
- Personalized styling AI

Verification

The Avatar Schema was verified through:

- Backend startup testing
- MongoDB schema validation
- Model export verification

Verified:

- Schema compatibility

- Backend integration readiness
- Database structure consistency

Conclusion

The Avatar Schema provides a scalable and AI-ready backend structure for personalized virtual try-on experiences in Raritone.

It prepares the platform for:

- AI avatar generation
- Virtual fitting systems
- Recommendation engines
- Future AR/VR integrations